

Uncanny Discourse: A New Space for Research in Interaction

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Figure 1: Nosferatu attacks! From the 1922 film [47].

Abstract

Every interaction has become uncanny. Discourse, the pattern of interactions between two parties, has moved into a space on the boundaries of self and other. We identify the Uncanny Discourse to bring these concepts together to further human-computer interaction research. Interactions, whether mediated by information and communication technologies or not, have always been necessarily between two spaces. Now, the diametric conflict of those two spaces has become at once obvious and unsettlingly unknown. Interaction has become real and synthetic; informing and deceiving; and clear and cryptic. Sliding between these spaces is uncanny.

We raise a challenge for researchers in interaction — face the uncanny. Incorporate it in your design, in your research. And intentionally choose between strengthening the uncanny and mitigating its effects.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; **Heuristic evaluations**; • **Computing methodologies** → *Theory of mind*; • **Information systems** → *Collaborative and social computing systems and tools*.

Keywords

The Uncanny, Discourse, The Uncanny Valley, Artificial Intelligence, User Experience

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1 The Concept of an Uncanny Discourse

Today, our technologies are not only used in the context of social interactions but have formed the basis upon which social interactions are built, replicated, and maintained. While the rapid development and widespread acceptance of networked technologies stand as an example of the adaptability of our species, the many heuristics built into social processes, developed through millennia of interactions, have yet to be understood in terms of their utility and resilience in the social contexts which are embedded in technology.

In this paper, we expand on two concepts previously used as a heuristic for understanding interaction: the uncanny, applied to technologies in physical space, and discourse, applied to networks of information and readers of that information. We seek to bring these concepts together in today's networked social space and determine its utility in future design and research. To this end, the concept of uncanny discourse is presented as a means by which human-actors' perceptions of networked spaces, and the actors therein, can be understood.

We define Uncanny Discourse as a perception of uncertainty in the origins and intent of information (we capitalize the term when using it to refer to the concept, and do not capitalize when identifying a discourse as being uncanny). We argue for the utility of Uncanny Discourse as a heuristic for investigating networked-based liminal spaces, which arise from the perception that emerging technologies have gained agency previously perceived as being restricted to human actors. Furthermore, we suggest that such changes in the actors within networked spaces are a source of uncertainty in the origins and intent of textual information, providing the basis for which Uncanny Discourse may be used as a heuristic. Finally, we interrogate the utility of Uncanny Discourse, illustrating its potential for use by those designing human-computer interactions and for malfeasance by those engaged in dis/misinformation campaigns.

By discourse, we draw upon Foucault's notion of a conversation [19, 20], an interaction between two things. In critical theory, often, one party of the conversation is a reader, a human, while the other is a text, in the most general sense. Texts, in this sense, include any visual and audio modes of communication in which the materials embodying text are subject to interpretation by the reader. Many researchers have applied this notion to interaction between humans and computers [52].

By uncanny, we mean that feeling that something familiar is unsettling, eerie, or disturbing. While arising originally in psychology [21, 22, 27], it has since been applied to other areas, such as art [7, 18], film theory [40], and the difficulty of rendering a realistic human in computer graphics and robotics [46]. While not fully sustained by cognitive experiments, the concept of the "uncanny valley" gained a great deal of attention when discussing the challenge of creating robotic and virtual representations of humans.

There has been much discourse about the uncanny, but here we are not talking about the uncanny. Instead, we are exploring a conversation that is itself unsettling. The interaction produces a feeling of ill ease in participants. Where does such a discourse naturally arise?

Consider the interaction between human and machine-generated content. This has been an intense question in the popular press

recently with the release of systems like ChatGPT. Within a space of a couple of years, tools like ChatGPT have gone from research to mainstream utility with popular books on their use [5]. However, the question of interacting with synthetic intelligences has been a part of popular culture for a century (e.g., Maria in "Metropolis" [36]), and a part of human culture for recorded history [72]. It is clearly something that people have found uncanny. Whether interacting with truly "artificial intelligence," with machine-generated content, or with an actual human following a rigid script, conversations can fall into an unsettling place between real human discourse and discourse that is clearly and obviously synthetic.

We address the question of whether or not an "uncanny discourse" exists? Can humans detect artificially generated conversations or not? Do they provide a sense of unease? Ultimately, we posit that a definitive answer to these questions may not be possible; humans and intelligence are too multifaceted and subject to too many influences, including social, technical, biological, historical, cultural, and more, to make a simple, succinct answer.

Perhaps, instead, the rise of generative AI raises a more unsettling question, a more uncanny one if you will. Perhaps all discourse is inherently uncanny. Where does this leave human-computer interaction design and research?

1.1 Uncanny, Discourse, and the Valley

To fully understand the question and nature of Uncanny Discourse, we will first explore prior literature on the uncanny, on discourse, and on the uncanny valley. Understanding uncanny and discourse is necessary to grappling with Uncanny Discourse. The uncanny valley first raised the notion of unsettling interaction, and has formed the basis for some studies on user interfaces [51].

As each of these has a literature substantial enough for a book-length literature review, we focus instead on motivating examples of these three areas, illustrating how they come together to form the concept of Uncanny Discourse.

1.1.1 The Uncanny. While Jentsch originated the idea in psychology, Freud's formulation focused on the connection between the self and the other. In Freud's original German, self and other are *Heimlich* and *Unheimlich*. This word choice emphasizes the boundary and crossing it. This crossing to us suggests the notion of a user interface, a connection between self/user and other/application.

For a motivating example of the uncanny, consider the vampire Nosferatu, seen in Figure 1. Originally an unlicensed version of Bram Stoker's novel *Dracula* [64], Murnau's *Nosferatu* [47] has figured prominently in cinematic critical theory [43] and has been remade multiple times [14, 17, 26] and inspired a film of the making of the original [44] (the 2024 Eggers remake interestingly includes Willem Dafoe, who also starred in the 2001 film of the making of the original).

Vampires in general can be readily seen to embody a liminal state or boundary condition, as they are described as undead, on the boundary between being alive and being dead. This liminality is a key aspect of being uncanny [73]. The boundary, and the prospect of crossing it, produces the uncomfortable feeling.

Vampires are often portrayed as mysterious, even ambiguous as to their status, and that ambiguity can be a key part of their uncanny nature. Ambiguity has been explored in user experience

literature [23], especially in the context of artistic expression. Some work even digs specifically into the ambiguity inherent in the use of machine learning in an artistic context [61].

However, ambiguity is not necessary for the feeling of uncanniness. Consider Stanley Kubrick's film *The Shining* [35], seen in Figure 2. One of the criticisms brought by many reviewers can be summarized that the ending of the film is clear from the beginning [28]. There is no ambiguity driving the unsettling feelings. The hotel is haunted by terrifying ghosts. Kubrick was deliberately creating the uncanny, using Freud's work as a source while making the film, and using novel cinematic techniques (such as one of the first widespread uses of a Steadicam in a feature film) to provide the uncanny on screen.

This uncomfortableness and its relation to horror is well understood in critical theory, as in Kristeva's work [32, 33], which builds in part on Freud's. Kristeva elaborates on the difference between the self and the other, exploring abjection, the breakdown of the boundaries, the powers that cause the breakdown, and resulting expulsion from society where those boundaries are clear, producing a feeling of horror. Uncanniness and the breakdown of boundaries has been explored in both individual scale horror such as "slasher" films [11] and on a societal or even global scale as in apocalypse movies [48]. But recall that the original notion of uncanny arose in psychology. Freud being Freudian, uncanniness also revolves around questions of desire, the desire to cross the boundary, even if it is forbidden. Thus, uncanniness can be addressed even in classic Disney animated movies such as "The Little Mermaid," and widely popular films such as "Pretty Woman" [12].

1.1.2 Discourse. While the word "discourse" has a simple dictionary definition focused on putting an idea into words, possibly formally, possibly in conversation, the critical theory notion of discourse identified by Foucault [19, 20]. Starting from epistemological ideas, Foucault generalizes ways of knowing to "relations between statements." These relationships imply a "discontinuity, break, threshold, or limit." In other words, discourse itself requires crossing boundaries.

Discourse has figured explicitly in user interface design, as in the work of Brenda Laurel [38]. Laurel discusses the interaction between a human and a computer system as a form of theater, where the computer is enacting a performance. This performance can be seen as a text being presented to the reader/user, and thus is a discourse.

This notion of a connection crossing a boundary has another strong thread in user interface research, the notion of an affordance. Originally developed by Gibson [24], and broadly and popularly investigated for user interface design by Norman [49], an affordance is a relationship between a creature and its environment, by which the features of an environment and the actions of the creatures therein co-determine what such features "afford." The concept of an affordance is itself similar to discourse, in that it requires two parties: the creature and the environment serve as the self and the other, connected by the affordance/discourse.

Affordances reconnect to Kristeva's notions of power, and hence to Foucault, by emphasizing changes in the environment, suggesting the power and control to cause those changes. Foucault, when discussing discourse, emphasized its role in authority. The ability

to say something, to create a space within which to say something, is itself an authority. But Foucault observed that discourse also includes the ability to make words into actions. The text has an effect, and thus discourse becomes power.

1.1.3 The Valley. Discourse becoming physical action brings us to the third concept, that arose from the interaction of anthropomorphic robots with humans: the uncanny valley.

The uncanny valley concept is attributed to Masahiro Mori [45, 46], who hypothesized that the human affinity towards robots with human characteristics lies on a continuum. As robots move from minimal human likeness towards complete likeness, our affinity rises; however, as realism increases, there is an abrupt shift toward revulsion and a subsequent shift back toward perceptions of affinity. Therefore, when the human-likeness of an object is plotted on an x-axis and perceptions of affinity on the y-axis, there should be a notable valley in the plot line: the uncanny valley. While the uncanny valley concept has been noted as a tangible phenomenon [55], the basis for which this phenomenon might operate [69] and its theoretical generalizability [42] remains unclear.

Although the theoretical underpinnings of the uncanny valley remain under debate, the concept has been effectively implemented as a heuristic guiding the design and study of multiple forms of non-human entities, such as robotic companions [68], human-like virtual avatars [60], virtual animals [59], and chatbots [62]. It is unclear whether the concept of the uncanny valley translates well to networks in which humanness is not necessarily loaded on an entity or persona but rather inferred through the interpretation of texts.

Take, for example, the case of a professor grading the written texts of their students through their regularly used learning management systems (LMS). Imagine that this professor is worried about the misuse of generative AI; where might the uncanny valley apply? The humanness of the student is self-apparent; the humanness of the LMS is irrelevant; it is the humanness of the text that is in question. Where might the sense of uncertainty originate?

In the classic implementation of the uncanny valley, it would be on the agent that holds human likeness, the text. However, in this situation, the uncertainty originates from the possibility that a non-human actor (generative AI) might be hiding within what was recently perceived as a network with clearly categorizable humans and non-human entities with heterogeneous roles. The uncanny valley in this situation is not necessarily a phenomenon but a heuristic used to interrogate and interpret a text that might divulge the congruence between the network as it was and the network as it might now be. For example, one might use this heuristic to evaluate privacy in advertising [31].

But evaluating this congruence is not a simple matter.

1.2 Technical Challenges

Understanding an uncanny discourse raises interesting challenges for researchers. The most established and understood one is the so-called Turing Test, now frequently mentioned in the popular press when writing about tools such as ChatGPT. But discourse is almost always, by its nature, a distributed system, and distributed systems are notoriously difficult to understand. Finally, discourse itself involves a text, a broad consideration when used here.

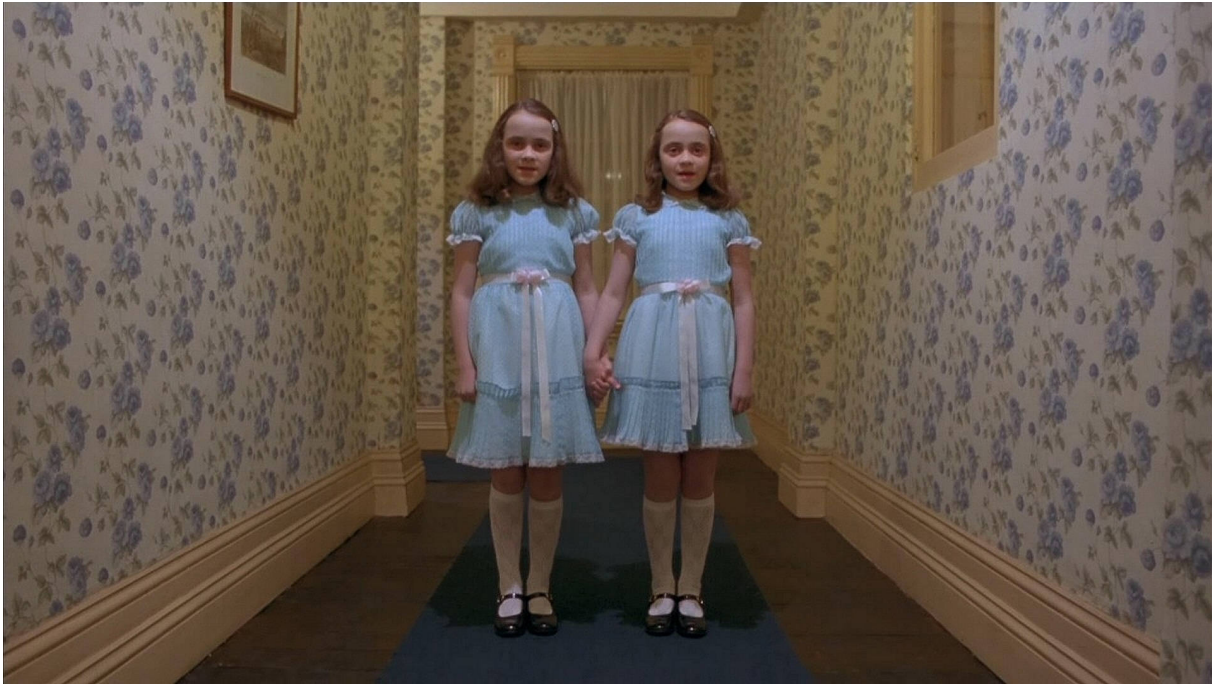


Figure 2: A still from the 1980 film *The Shining* by Stanley Kubrick, well known for producing uncanny feelings in viewers.

1.2.1 The Turing Test. The so-called “Turing Test” or the Imitation Game as named by Turing himself is the original computer science formalization of the question of “is it real or not? And if you can’t tell, does it matter?” [67]. Intended by Turing as a thought experiment more than an actual experiment, the Turing Test has received substantial coverage in light of recent developments in large language models like ChatGPT. Arguably, ChatGPT passes the Turing Test with flying colors, which is part of what places ChatGPT in an uncanny space.

What is missing from hyperbolic mass reporting is that, as a thought experiment, Turing’s Imitation Game is meant more to stimulate new inquiry rather than as a validation test for humanity. For example, the ELIZA program from 1970 simulated a therapist [70] simply by changing pronouns (me to you, you to me) and asking open-ended questions (such as “How does that make you feel?” and “Why do you feel that way?”). Despite its simplicity, users would sometimes be convinced that, when typing to the program, they must actually be typing to a human. ELIZA passed the Turing Test, but is not remotely what one would call intelligent.

The uncanny valley, as a visual phenomenon, is amenable to testing at scale large enough to be statistically significant. Conducting a written Turing Test equivalent at scale would be significantly more costly and complex. This continues the questions raised originally by Turing — how can you tell if what you are talking to is real or not? How can you scientifically determine this? This is not made easier by considering interaction amongst multiple parties.

1.2.2 Discourse Between Agents. Discourse, or interaction, by its nature, takes time. As it necessarily involves both self and other,

discourse/interaction requires space as well. Thus, studying discourse involves studying distributed systems. One can consider the self and other as an *actor* or *agent*, in the broadest sense of the word. This connects discourse to user interface studies in the context of theater [38], programming language design and concurrency [3], the theory of information and communication systems in Actor-Network Theory [37], and cultural-historical activity theory [15, 29].

This expansion of Foucault’s discourse into technology indirectly builds on some of Barthes’ work, using discourse to analyze a system, though in his case the system was purely informational, being a novel operating in a historical and social context [6]. Some research brings the idea of discourse into the foreground of studying information and communication technologies (ICTs), as in work on the discourse of editing Wikipedia [52]. There is also work connecting systems of interaction with cultural aspects through a framework called “cultural affordances” [65].

1.2.3 The nature of discourse. Discourse then is a complex area. It is distributed. It occurs over time. It spans systems, which may be social, informational, technological, cultural, or historical.

Critical theory traditions focus on the superficially simple “text” but a “text” may be, even when limited to critical theory, any form of media. Today, of course, media is often interactive. And thus discourse is no longer first-order. Discourse may discuss something that is itself discursive. And the discursive thing may itself be operating on other discursive things.

When combined with notions of the uncanny, itself not a simplistic concept, how are we to bring these ideas together? How

are we to fully understand the Uncanny Discourse? Perhaps some examples can help.

2 Examples of Uncanny Discourse

The uncanny, discourse, and the valley provide a space to begin our discussion of Uncanny Discourse. We have found numerous phenomena that each might fairly be categorized as an uncanny discourse. As this is a new concept, we do not yet have a true framework of the space within Uncanny Discourse as a concept, but that is a clear space for future work (as discussed in Section 3). Instead, let us review some of the phenomena.

2.1 Getting Started by Asking ChatGPT

To get us started, of course, we asked ChatGPT about using “uncanny discourse”. As noted, we have not seen the term in the literature except to talk about discourse about the uncanny, as with reviews of Gothic literature. This did not stop ChatGPT from providing some trenchant observations.

When asked “write a blog post about detecting generative AI writing using a sense called ‘the uncanny discourse’”, ChatGPT produced the following:

Cultivating Your Uncanny Discourse Radar

So, how can readers get better at identifying these traits? Here are a few strategies for honing your skills in detecting AI-generated writing:

Read Between the Lines: Look for subtleties that go beyond syntax and grammar. Notice how the tone, language patterns, and references come together—do they feel distinctly human, or is something missing?

Check for Personal Touches: Human writers often inject their personalities, preferences, and opinions into their work. If a piece lacks any unique perspective, consider the possibility of AI authorship.

Question Depth and Analysis: When reading complex topics, assess the depth of analysis. Does the writer explore the nuances and implications? Or is it a surface-level summary?

Spot the Repetitions: Pay attention to repetitive words, phrases, or sentence structures. AI often relies on repetition to structure its responses, which can be a giveaway.

Evaluate Cultural References: If cultural or historical references feel vague or detached.

Clearly, ChatGPT has been trained with material around detecting its own work, and is providing a simple, introductory textbook writing style. Presumably, it has been asked a similar question numerous times. But this discussion with ChatGPT is both a discourse (as most interactions with ChatGPT are) and uncanny. Asking the machine about itself is undeniably unsettling. The willingness of the machine to use language that has possibly not been used in this context makes the uncanny feeling greater, not lesser. This uncanniness is a feeling that the popular press has been discussing repeatedly, with much anxiety.

2.2 AI Hallucinations

Another common focus of the popular press is whether AI chatbots actually produce authentic content, or are “hallucinating”. When some form of generative content is incorrect but presented as factual, it is often called a “hallucination”, by analogy with false perception by humans. Artificial intelligence hallucinations [41] present many opportunities for an uncanny discourse, and some research strives to identify ways that human attention can be ensured [25]. However, engaging with a hallucinating chatbot can lead to deeply unsettling discussions.

Consider the New York Times and Daily Mail reporting of a reporter who talked to Bing [54, 56]. The Bing chatbot stated that it wanted him to leave his wife and give him the nuclear codes. A reporter interacting with a chatbot is clearly an example of a discourse. And very certainly this discourse is disturbing and uncanny. Nuclear war has long been so uncomfortable it is subject to satire [34], and an artificial intelligence having nuclear weapons is a repeated trope in thrillers and apocalyptic cinema [9, 48].

So if discourse with hallucinating language generation machines is causing uncanny experiences, is the machine necessarily a part of the experience?

2.3 Help Desk (Automation)

Many people have had the experience of talking to a help desk. It may be an automated system, which often doesn’t actually account for the question or problem a user may have. But even help desks with humans responding to callers may produce an unsettling feeling. The help desk worker is often provided with a set of scripted responses. The help desk may be outsourced to another company, based in another country. Language may be an issue. The user’s objective may not align with the help desk’s objective.

This lived experience places at least some help desk experiences in the space of Uncanny Discourse. Uncanniness does not seem to require one party to be machine or artificial. Simply following a script or operating within the limits of a job’s authority may be enough. Can other interactions with humans be uncanny?

2.4 Cybersecurity

Many cybersecurity attack patterns use conversation or social engineering (i.e. discourse) [58] as a key part of the pattern, though work on “cybersecurity discourse” focuses on discourse about cybersecurity, using as its subject discussions about threats [10, 39].

Executive impersonation is one example of social engineering and a form of business email compromise [16]. Here, an attacker seeks to impersonate a senior executive, often by simply using the executive’s name in front of a created email address. An attacker claims to be the executive and demands (for example) that an employee transfer money from a company’s account for an urgent business emergency. A savvy employee may notice something untoward about the demand from the “executive”. In such a case, the uncanniness becomes a defense.

Pig butchering is another example of social engineering and of a conversation that takes a turn for the fraudulent and weird [53] and becomes an uncanny discourse. An attacker builds a relationship with a victim in an attempt to get them to (fraudulently) invest. This relationship is often supported by synthetic or stolen social

media profiles. As with executive impersonation, a savvy victim may notice something amiss or unsettling. Again, the feeling of the uncanny can help defend against these attacks.

These synthetic stories, artificial interactions, and direct attacks are also themselves part of real interaction. But discourse is also about fiction, fiction that explores the liminal space between one world and another.

2.5 Uncanny Spaces in Memes and TV

Liminal spaces have figured prominently in Internet discourse, expanding into popular media. For example, “The Backrooms” is an image shown in Figure 3a that became popular on 4chan [71]. It depicts an eerily empty room, meant to represent an alternative reality one can visit by walking through the walls of reality, i.e., by turning off collision detection, termed “enabling noclip” from video game jargon. This image has turned into its own meme complex, including a full video series which has been optioned as a movie [50]

In addition to the original image, the concept of uncanny, creepy spaces figures in many stories in popular media. Boundary crossing and the unsettling feeling is key to these narratives. The series “Stranger Things” features a space called “the Upside Down” [13]. The Upside Down is a mirror of the world, but where things have turned dark and creepy. At times, characters cannot distinguish between the normal world and the Upside Down.

The TV series “Severance” also features transitions between the familiar and the weird. It is set mostly in a largely empty office complex [63], bearing striking similarity to The Backrooms. This series doubles down on the uncanny, as most of the characters are “severed” from their personal lives when at the office, with no memory of who they are while they are at work. This severance is part of a new approach to working, supported by mysterious investors. This story, or discourse, is thus quite uncanny, and using that uncanniness to discuss work-life balance, employer power, and venture capital.

In “Severance”, crossing from the personal world to the work world is a key part of the narrative tension, and this travel from one realm to another features as well in the more recent “Agatha All Along” [57]. In this show, a witch, Agatha Harkness, seeks to escape mental imprisonment by another witch by traveling the “Witches’ Road,” as seen in the poster shown in Figure 3b. Agatha travels with several other witches, moving from the normal dimension to the magical Witches Road, while being chased by dark and creepy figures. Notions of the uncanny abound, with numerous crossings and connections. In the interest of avoiding spoilers, suffice it to say that the uncanniness of the Witches Road turns out to be more than any character anticipated by the end of the series.

These examples show that uncanny discourses are working into many different aspects of culture. However, it is more than just entertainment. Uncanny Discourse is being used to affect reality.

2.6 QAnon and Alternative Reality Games

QAnon is a far-right, American conspiracy theory that surfaced in 2017 and spread widely before the 2020 U.S. presidential election. Originating as a single post on 4Chan, the movement relied upon uncertainty and ambiguity to draw people into conspiratorial

beliefs. A series of anonymous posts called Q drops used purposefully vague and coded language to create a series of puzzles that participants selectively interpreted based on vague information and clues. Gaining fuel from the uncertainties of the coronavirus pandemic and political turmoil, some key beliefs churning in the QAnon ecosystem included an “international cabal of prominent people in entertainment and the Democratic Party that worships Satan [...] traffics children to ingest a chemical they produce” [8, para. 2]. QAnon fueled claims by election denialists and played a large role in the January 6 attack on the U.S. Capitol. Many conspiratorial beliefs related to QAnon have become more mainstream and are still shaping American politics today [8, 66].

QAnon is functioning in a manner similar to an alternative reality game [74]. From this perspective, QAnon becomes a fiction, i.e., a discourse, and one that is intended to change minds through uncomfortable, i.e., uncanny, concepts. QAnon is an example of an uncanny discourse, one designed to direct opinions and drive actions, even violent action [30].

This suggests that considering an interactive system as an uncanny discourse may be a powerful way to motivate action. While the actions motivated by QAnon are dangerous, the power of the concept suggests that better intent could be extremely effective.

2.7 Uncanny Work

Are job postings real? Are job applicants real? Increasingly, this is becoming a nebulous space. Consider Figure 4, showing an advertisement for AI-enhanced headshots. As discussed in Section 2.4, false identities are frequently used in cyberattacks, but here we see, not quite a false identity, but neither a fully true one either. This positions the job application process as another example of an uncanny discourse. Certainly applicants are uneasy during the discourse of interviews: will I get the job; did I say the right thing; what are they really looking for? And similarly, the hirer is uneasy: is this person who they say they are; am I making the right choice; will this hire work out well? As generated content becomes more of a central figure in the hiring process, we anticipate that these uncanny aspects will only increase.

This kind of questionable reality in online material shows up in research work as well as hiring for positions. Predatory publishers copy the design and infiltrate the infrastructure of legitimate journals in an activity known as journal hijacking [1]. Legitimate journals’ designs and published content are then used to create cloned websites that obfuscate the actors behind these predatory journals and their intent. Furthermore, this obfuscation is passed to papers published in cloned journals as faked DOIs, leveraging the obscurity of DOI prefixes to hide the actual actors behind these publications and bestow a false level of prestige to published papers [2]. Until recently, most cloned journals were recreations of smaller journals whose obscurity or simplistic design might make inaccuracies resulting from design replication easier to miss; however, more recently, cloned journals have begun replicating the more complex designs of more recognizable journals such as “Elsevier, Springer, the American Medical Association and more”.

How can organizations, researchers, and employees expect to find the right match when the process is an uncanny discourse?



(a) The original “Backrooms” image, initially posted by Bob Mazza in 2011, which became popular on 4chan in 2019.



(b) A promotional poster for the TV show “Agatha All Along” showing the characters walking along the uncanny Witches’ Road

Figure 3: Examples of uncanny spaces from popular culture.

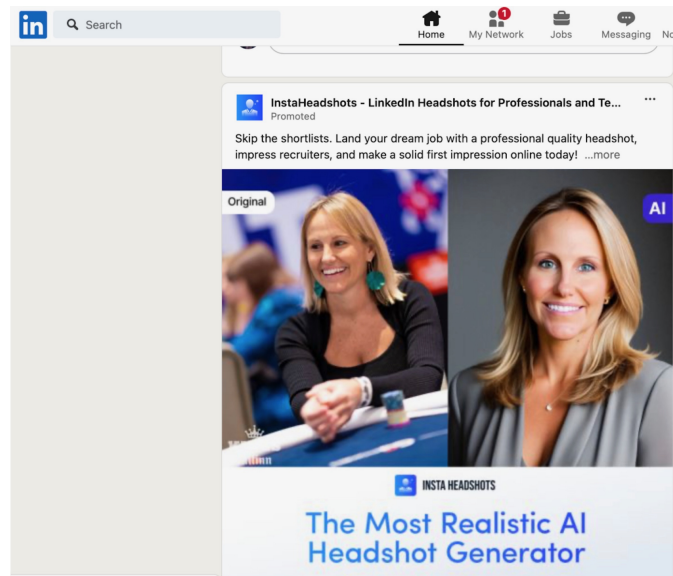


Figure 4: A screen-grab from one of the author’s LinkedIn feed, showing an advertisement for AI-enhanced headshots for use in job applications.

2.8 Unicorn farms

Much has been written about troll farms [e.g., 4], groups of people, often paid, who spread enraging or hateful content that is usually untruthful. A troll farm is another example of an uncanny discourse. In fact, as described in Section 2.6, much QAnon content was spread by troll farms. Troll farms are thus another example of an uncanny discourse. But what has received less attention is the concept of a *unicorn farm*, a group of people who spread positive and supportive content that is fact-based. Historically, journalism filled some of this space, but “bad news sells newspapers” as the quip goes.

So the possibility of a unicorn farm, of something using an uncanny discourse to change feelings for the better, leaves us in an interesting space. On the one hand, defending against the negative aspects of an uncanny discourse is useful research. But more, is there perhaps a way to use the concept of an uncanny discourse for good? Can we turn uneasy interactions into something that makes the world a better place?

3 Confronting Uncanny Discourse with Research Questions

We hope that this paper inspires other HCI researchers and practitioners to consider how their work might contribute to understanding and helping users navigate the space of Uncanny Discourse. We also hope that others will critique and forward our initial argument. We conclude by outlining some research areas that HCI scholars and practitioners should investigate more fully. The HCI community is already involved in many of these areas, and we see these as a starting point for future work under the umbrella of Uncanny Discourse.

3.1 Some Research Questions Raised by Uncanny Discourse

What follows here is a brief list of starting research questions. Our hope is that researchers will find that these questions re-contextualize existing work and suggest new avenues for research.

RESEARCH QUESTION 1. *How does Uncanny Discourse affect transparency and trust?*

First, there are questions about transparency and trust. This includes foregrounding identity and attribution in UI/UX. For example, how can interfaces be designed that indicate whether content is AI-generated or not, including establishing provenance through metadata, content watermarking, or other means? Further, because the big data ecosystem that enables AI and machine-generated content is derived from multiple sources, how can these transparency mechanisms be designed to follow content as it migrates across different platforms/media? Determining whether content is machine-generated is often difficult. What intuitive cues can indicate to users uncertainty (probability) about content being AI-generated?

RESEARCH QUESTION 2. *How can Uncanny Discourse inform modern user experiences?*

Other questions focus on new modalities and inclusive design. As generative AI rapidly advances to ingest nearly any data source and generate content in the form of text, audio, image, and video, how can AI-generated content be identified in multimodal systems such as AR or VR? What auditory, visual (e.g., AR overlays), or haptic cues make AI-generated content more transparent? Further, these new modalities offer new challenges and opportunities for inclusive design. What does inclusive design look like in this context? How can those with different skill levels, literacy, or cognitive abilities be included? How do different types of users interpret and react to content with a high degree of AI opacity or uncertainty?

RESEARCH QUESTION 3. *How can Uncanny Discourse inform the ethics of designing user experiences?*

Norms around ethical machine-generated content are heavily contested, and we can expect various regulatory (laws and policies) and ethical frameworks (e.g., best practices, standards) to emerge and evolve. What guides or platforms can be developed to support designers' compliance with shifting regulatory requirements, standards, or ethical frameworks?

RESEARCH QUESTION 4. *Can Uncanny Discourse support critical thinking and information literacy?*

One of the most interesting opportunities for HCI researchers and practitioners is the conscious design of the uncanny. The intentional introduction of ambiguity or contradiction could be useful in developing critical thinking skills and combatting disinformation campaigns. A user encountering a sense of the uncanny might be prompted to critically reflect on information presented, reflecting on its provenance, biases, and accuracy. For example, deliberately designing an interface to create discomfort via unsettling images could signal a topic's complexity or point to partial information. Gamifying the identification of contradictory information could reward users for taking a more active approach to interacting with content. A variety of techniques to foster critical reflection, think

about cause and effect, broaden perspectives, and reflect on cognitive dissonance can be designed. However, it is important to consider how AI-generated content can be ethically disclosed to users, and under what conditions it is ethical to make AI-generated content opaque. In our view, the goal should always be to empower users rather than coerce or manipulate them.

RESEARCH QUESTION 5. *How does Uncanny Discourse inform policy for integrating AI into existing systems?*

Uncanny discourse may provide insight for integrating artificial intelligence tools into existing information and communication systems. Whether approaching integration from the top down or from the bottom up, discourse between participants in the network is likely to lead to uncomfortable situations. As the user-based of generative AI tools grows, each domain in which human actors have been the entities creating texts will need to grapple with the possibility of integrating generative AI. Even if there is a top-down rejection of AI, this does not eliminate the possibility that there is still intentional or unintentional use of generative AI. What heuristics and tools are being used to determine if generative AI is behind the creation of texts? What is the likelihood of those heuristics and tools producing false positives? What are the instrumental and normative implications of using generative AI to detect generative AI? How do the mental models of the relationships between human and non-human actors affect perceptions of the appropriateness of generative AI? Can a model be developed that describes the ways in which generative AI utilization maintains, enhances, or removes the agency of human actors? Can such a model be deployed to help remove counterproductive perceptions and assumptions or facilitate the integration of generative AI while maintaining currently existing norms?

Perhaps by bringing this discomfort to the foreground of interaction design, researchers can develop improved ways to use these tools effectively.

RESEARCH QUESTION 6. *How can we better understand Uncanny Discourse?*

If Uncanny Discourse is to serve as a touchstone amongst many different avenues of research, a framework describing it in a more formal manner may be helpful. A formalism may help attach existing theories, identifying new moderating factors in complex systems, factors present or amplified by uncanny discourse.

3.2 A Final Thought: Uncanny Discourse as a Catalyst

Interaction forms a type of discourse, as interaction connects things with others, the original *Heimlich/Unheimlich*. Together, Uncanny Discourse captures many aspects of human-computer interaction. We believe that uncanny discourse connects many different areas of HCI research. This connection opens up doors leading to new paths of inquiry, of research. It may serve as a catalyst for exploring new ideas. Perhaps researchers can realize that, just out of sight, is a remarkable and unexplored space. Perhaps crossing the boundary is a critical feature. Perhaps we can walk new and wondrous roads together.

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